Organic Cumulative Examination

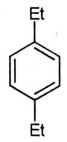
September 7, 2018

This Cume has 20 points and 14 points or more are needed to pass

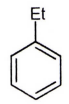
1) Describe an absolute necessary condition for a nucleus to allow measurements by NMR spectroscopy (2 points)

2) Examine the following two compounds and state the number of ^{1}H and ^{13}C NMR signals

(a) 2 points

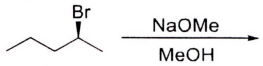


(b) 2 points

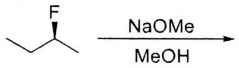


3) Complete the following reactions and give the major organic reaction product(s) (2 points each)

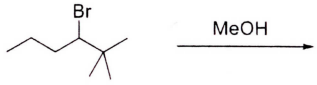
(a)



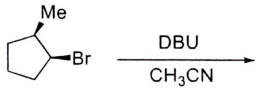
(b)



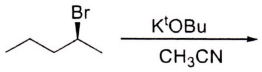
(c)



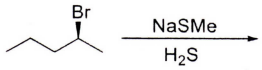
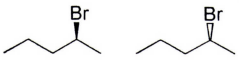
(d)



(e)



(f)



(g) You are given the two compounds below and are supposed to assign unambiguously, which of the two compounds is R and which one is S by ^{1}H NMR spectroscopy (the polarimeter is broken☹). Which experiment would you propose? What would you do? How do you get the needed information? (2 points)